

RISHABH PRATAP SINGH

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Masters Student (Computer Science & Software Engineering)

TECHNICAL SKILLS

Languages - C++, Python, Go, Zig, TypeScript, SQL

ML & HPC: PyTorch, CUDA, MPI, OpenMP, Nsight, RAG, OpenCV

Technologies - ReactJS, Spring, FastAPI, AWS, Terraform, REST APIs, PostgreSQL, Snowflake, NoSQL

EDUCATION

Master of Science in Computer Science from **University of Washington** Washington, US - **GPA - 4.0/4.0**

2024-Present

- **Coursework** - Parallel Programming, High Performance Computing, Deep Learning, Computer Architecture

Bachelor of Technology in Computer Science from **Poornima University** Rajasthan, India - **GPA - 8.37/10.0**

2016-2020

- **Coursework** - Data Structures, C, OOP, Software Engineering, Web Technologies, DBMS, Networking

WORK EXPERIENCE

Research Assistant, University of Washington Washington, US

September, 2025 - Current

- **Extended MASS C++/CUDA (high performance agent-based modelling library)** by implementing functional programming primitives, asynchronous inter-node computation, and graph-based computation on a cluster.
- Designed and benchmarked graph algorithms (BFS, PageRank) under multiple partitioning strategies, improving throughput by **20%** on a **24** node cluster.

Data Engineer/Full Stack Developer at Wipro(Blackstone Account) Haryana, India

March, 2022 - September, 2024

- Led the development of a **highly dynamic Cash Flow Dashboard** (React + FastAPI), used to create and track settlement requests and their transactions. It provides an **intuitive UX for Finance Teams** for keeping a centralized record.
- Built user-customizable integrations with Blackstone Data Warehouses, that **fetches and updates relevant transactions saving at least 50 person hours per week**.
- Engineered a **change tracker** for full audit history of financial records, **improving compliance and accountability**.
- Applied **Fuzzy matching algorithm** to find related documents from Data warehouse, facilitating a proper **searching mechanism**.
- Implemented a **rule engine** for automated report generation, saving **20+** person-hours per week.
- Deployed a private DLT (Distributed Ledger Technology) architecture for financial information sharing/logging.

Digital Specialist Engineer at Infosys Ltd. Karnataka, India

February, 2021 - March 2022

- Built full-stack features and RESTful APIs, integrating microservices with AWS Lambda via event-driven architecture.
- Developed services for dynamic interaction with third party payment and shipping services.

PERSONAL PROJECTS

Memory Graph

May 2026

- Designed a hierarchical graph memory for LLM agents using semantic and temporal edges and PageRank-style importance weighting, improving context recall.
- Built nested subgraph summarization that cut context tokens.
- Implemented a semantic response cache (similarity threshold and staleness expiry).

Simple Encrypt [Website Github](#)

January 2026

- Built a zero-backend browser/terminal encryptor for text and files using AES-256-GCM with optional expirational metadata, deployed via GitHub Pages.
- Tech Stack: **HTML/CSS/JavaScript, Web Crypto API (AES-256-GCM), Github Pages for deployment, Go (Terminal ver)**

N-Body Simulation [Github](#)

January 2025

- Implemented three N-Body simulation algorithms (Barnes-Hut $O(n \log n)$, direct summation $O(n^2)$, and Fast Multipole method $O(n)$) with configurable scenarios and benchmarking support.
- Tech Stack: **C++, CUDA, OpenCV, CMake**

CERTIFICATES

Bootdotdev Backend Developer Path - [Boot.dev](#) (Archmage) (Go, HTTP Server, HTTP Protocol, Cryptography)

February 2026

Algorithmic Toolbox - University of California San Diego (Coursera)

August 2019